

CyberArk Practice Tracker — Documentation Suite

PART 1: EXISTING FUNCTIONALITY RECORD

A plain-language inventory of every feature currently in the app.

1.1 App Structure

The app is a single HTML file with no backend. All data lives in JavaScript variables in memory and resets on page refresh. The layout has three sections:

- **Left sidebar** — navigation between the three main pages
- **Top bar** — page title, subtitle, and today's date
- **Main content area** — changes based on the active page

There are three pages accessible via the sidebar:

1. **Dashboard**
 2. **Projects**
 3. **Customers**
-

1.2 Dashboard Page

Shows a summary of the entire practice. Contains:

- **Five stat cards** — total projects, count by type (Implementation, Managed Service, License Renewal, New Opportunity)
 - **Needs Attention panel** — lists all projects whose status is "At Risk" or "Delayed"
 - **Due in 30 days panel** — lists projects whose due date falls within the next 30 days
 - **All Projects mini-table** — a read-only table showing every project with: customer name, project type, owner, progress bar, status badge, and due date. Clicking any row navigates to the Projects page.
-

1.3 Projects Page

The main working view for tracking project tasks.

Filters available:

- Tab bar to filter by project type (All / Implementation / Managed Service / License Renewal / New Opportunity)
- Dropdown to filter by project status
- Dropdown to filter by owner

Project list: Each project shows as a row: customer name + notes, type badge, owner avatar + name, progress bar (% of sub-tasks marked done), status badge, and due date. Clicking a row expands or collapses a task panel below it.

Adding a new project: Clicking "+ Add Project" opens a modal with these fields:

- Customer name (required)
- Project type (dropdown: Implementation / Managed Service / License Renewal / New Opportunity)
- Owner (free text)
- Status (dropdown: On Track / At Risk / Delayed / Completed / Prospecting)
- Start date
- Due date
- Notes

On save, the app auto-generates a task list for the project based on which project type was chosen (see templates below).

Editing a project: The "Edit project" button inside the expanded task panel opens the same modal pre-filled with existing values. All fields are editable.

Task templates (auto-generated per project type):

Implementation — 17 activity groups including: Discovery & Scoping, Infrastructure Deployment, Vault Configuration, Safe Management, Account Onboarding, CPM Integration, PSM Configuration, MFA Setup, LDAP/AD Integration, Reporting & Analytics, Disaster Recovery, Command Control, Clipboard Control, Hosted Apps SSO, Session Recording, Documentation, Training.

Managed Service — 5 groups: Service Onboarding, Monthly Health Review, Incident Management, Platform Updates, Reporting.

License Renewal — 5 groups: Renewal Discovery, Commercial Proposal, Negotiation & PO, Renewal Execution, Documentation.

New Opportunity — 4 groups: Opportunity Qualification, Pre-Sales & POC, Proposal & SOW, Deal Closure.

Task panel (expanded project view):

- Shows total sub-task completion count (e.g., "3/12 sub-tasks completed")
 - Button to open the Edit Project modal
 - Button to jump to that customer's profile in the Customers page
 - Each activity group (e.g., "Vault Configuration") is collapsible
 - Each group shows its own completion ratio and mini progress bar
 - Inside each group, sub-tasks are listed. Each sub-task has:
 - Name (strikes through when done)
 - A date picker to set a due date for that sub-task
 - A checkbox to mark it done/undone
-

1.4 Customers Page

A two-panel layout: customer list on the left, customer detail on the right.

Customer list:

- Alphabetically sorted
- Search box to filter by name
- Each item shows: customer name, project type, owner name, status dot
- Clicking selects the customer and loads their detail

Customer detail — four tabs:

Overview tab:

- Project status section: type, owner, status, progress bar, start date, due date
- Customer profile section: industry, CyberArk tenant URL, region/DC, identity provider, SIEM, license type, license count, license expiry
- Notes section (if any)
- "Edit profile" button opens an inline modal to edit all profile fields

Contacts tab:

- Lists all contacts as cards, each showing: initials avatar, name, role, department, email, phone, notes
- "+ Contact" button opens a modal: name (required), role, department, email, phone, notes
- "Remove" button deletes a contact

Documents tab:

- Lists documents grouped by type: HLD, LLD, SOP, Architecture Diagram, Network Diagram, Infra Details, Test Plan, UAT Sign-off, Other
- Each document row shows: icon, name, notes, status badge (Draft / In Review / Approved / Shared)
- Clicking a row opens the document link in a new tab (if a URL was provided)
- "+ Document" button opens a modal: name (required), type (dropdown), status (dropdown), link URL, notes
- Delete button on each document row

Infra tab:

- Lists servers grouped by environment: Production, UAT, Dev, POC
- Each server shows: hostname, IP, OS, role badge, notes
- "+ Server" button opens a modal: hostname (required), IP, OS, role (PVWA / CPM / PSM / DR Vault / Primary Vault / Jump Server / Other), environment (dropdown), notes
- Delete button on each server row

1.5 Project Status Values (current)

Status	Meaning
On Track	Project progressing as planned
At Risk	Some risk of delay or issue
Delayed	Overdue or significantly behind
Completed	All work done
Prospecting	Early stage, not yet active

1.6 Data Model (current, in-memory)

Project:

- id, customer, type, owner, status, start, due, notes, tasks[]

Task group (inside project.tasks[]):

- name, open (collapsed/expanded), subs[]

Sub-task (inside group.subs[]):

- name, done (boolean), due (date string)

Customer profile:

- contacts[], documents[], infra[], profile{}

Profile fields:

- industry, tenant, region, idp, siem, licenseType, licenseCount, licenseExpiry, notes
-
-

PART 2: PRODUCT REQUIREMENTS DOCUMENT (PRD)

Product: CyberArk Practice Tracker — Frontend Enhancement v2

Prepared for: Engineering / Frontend

Status: Ready for Implementation

2.1 Problem Statement

The current tracker gives a good high-level view of projects but lacks task-level accountability. Managers cannot tell who is responsible for a specific step, what is blocking a step, or why a project is delayed. The only person tracked per project is the overall "owner", making it impossible to use the tracker for day-to-day task assignment and follow-up.

2.2 Goals

1. Allow any sub-task to be created, edited, or deleted — not just completed or have its date changed.
 2. Assign a specific person (assignee) to each sub-task, and give any user a view of tasks assigned to them.
 3. Give each sub-task a meaningful status that explains where it stands — not just "done" or "not done".
 4. Allow a sub-task to be flagged as blocked, waiting for a resource, or waiting for someone to respond.
-

2.3 Non-Goals (out of scope for this version)

- Backend / persistent storage (still in-memory)
 - Authentication or user accounts
 - Email or push notifications
 - Mobile-specific layouts
 - Reporting exports (PDF / CSV)
-

2.4 Feature Specifications

FEATURE 1 — CRUD for Sub-tasks and Activity Groups

What it is:

Users can add a new sub-task to any activity group, edit the name of an existing sub-task, delete a sub-task, add a new activity group to a project, edit the name of an existing activity group, and delete an activity group.

Trigger locations:

- Inside any expanded activity group: a small "+ Add step" link appears after the last sub-task row.
- Each sub-task row gets an edit icon (pencil) and a delete icon (×) on hover.
- Inside the task panel header area: an "+ Add phase" button adds a new activity group.
- Each activity group header gets an edit icon and a delete icon on hover.

Behaviour — Add sub-task:

1. User clicks "+ Add step" inside a group.
2. An inline input row appears at the bottom of that group's sub-task list.
3. User types the step name and presses Enter or clicks a confirm checkmark to save, or presses Escape to cancel.
4. The new sub-task is added with: name as typed, done = false, due = empty, assignee = empty, status = "Not Started".

Behaviour — Edit sub-task name:

1. User clicks the pencil icon on a sub-task row.
2. The sub-task name text turns into an inline editable input, pre-filled with the current name.
3. User edits and presses Enter to save, or Escape to cancel.

Behaviour — Delete sub-task:

1. User clicks the × icon on a sub-task row.
2. A confirmation prompt appears inline ("Delete this step? Yes / Cancel").
3. On confirm, the sub-task is removed from the array and the panel re-renders.

Behaviour — Add activity group:

1. User clicks "+ Add phase" in the task panel header.
2. A small modal or inline input appears asking for the group name.
3. On save, a new empty group is added at the bottom of the project's task list.

Behaviour — Edit activity group name:

1. Pencil icon appears on the group header row on hover.
2. Clicking opens an inline input with current name.
3. Enter to save, Escape to cancel.

Behaviour — Delete activity group:

1. × icon on group header on hover.
2. Confirmation prompt: "Delete this phase and all its steps? Yes / Cancel".
3. On confirm, the entire group including all its sub-tasks is removed.

Edge cases:

- Deleting the last sub-task in a group should leave the group visible but empty (not auto-delete the group).
 - Sub-task names must not be blank — show an inline validation message if the user tries to save an empty name.
-

FEATURE 2 — Per-Sub-task Assignee and "My Tasks" View

What it is:

Each sub-task can have one assignee chosen from a shared list of team members. A new sidebar page called "My Tasks" shows all sub-tasks across all projects that are assigned to a chosen person.

Data changes:

- Sub-task object gains one new field: `assignee` (string, default empty)
- A top-level `teamMembers` array stores the list of known people (starts pre-populated with the existing project owners: Rahul, Priya, Amit, Sneha)
- Team members can be added or removed from a settings area (see Feature 2b)

Sub-task row change:

The sub-task row layout changes from:

`[name] [date picker] [checkbox]`

to:

`[name] [assignee chip] [date picker] [status badge] [checkbox] [edit/delete icons]`

The assignee chip shows the person's initials (coloured avatar, same style as existing owner avatars). Clicking the chip opens a small dropdown to pick a person from the team list, or clear the assignment.

My Tasks page:

- Added as a new nav item in the sidebar below "Customers"
- At the top: a person-picker control (dropdown or tab strip showing all team members)
- When a person is selected, the page lists all sub-tasks assigned to them, grouped by project
- Each item shows: project name (customer), activity group name, sub-task name, status badge, due date, and whether it is done
- Items are sorted: overdue first (past due date, not done), then due soonest, then no due date, then done
- Clicking a project name expands to show full task panel (same as Projects page)

Feature 2b — Manage team members:

- A small gear icon in the sidebar footer opens a "Team" management panel (modal or drawer)
 - Shows current team members as a list
 - "+ Add member" opens an inline input for a name
 - Each member has a remove button (disabled if they are currently assigned to any open sub-task — show tooltip explaining why)
-

FEATURE 3 — Sub-task Status

What it is:

Each sub-task has a status field that replaces the binary done/not-done checkbox with a richer state. The checkbox remains as a shortcut for marking something complete, but the full status is selected from a dropdown.

Status values:

Status	Colour	Meaning
Not Started	Gray	Work has not begun
In Progress	Blue	Assignee is actively working on this
In Testing	Teal	Work is done, verification/testing is underway
Awaiting Feedback	Amber	Waiting for a response, review, or approval from someone
Blocked	Red	Cannot proceed — see flag reason (Feature 4)
Done	Green	Fully complete

Rules:

- When the checkbox is clicked and turned on, the status automatically changes to "Done".
- When the checkbox is un-clicked (un-done), the status automatically goes back to "In Progress".
- The user can also click the status badge directly to open a small dropdown and pick any status manually.
- If a status is set to "Done" via the dropdown, the checkbox also becomes checked.

- If a status is set to anything other than "Done" via the dropdown, the checkbox is unchecked.

Impact on progress bar:

The progress percentage formula changes. Instead of counting only `done === true`, it counts sub-tasks where status is "Done". The calculation stays the same: $\text{done count} / \text{total count} \times 100$.

Impact on dashboard:

The "Needs Attention" panel can now optionally highlight projects that have one or more sub-tasks with status "Blocked" or "Awaiting Feedback", even if the overall project status is "On Track". A small indicator (e.g., a count badge "2 blocked") appears next to the project entry.

FEATURE 4 — Block / Flag a Sub-task

What it is:

A sub-task can be flagged to explain why it is stuck. The flag captures: the type of blockage, a short description, and optionally who or what is causing the wait.

Flag types:

Flag Type	When to use
Waiting for resource	A server, licence, environment, or tool needed for this step is in use by another process
Waiting for approval	Needs sign-off from a person or team before it can proceed
Waiting for customer	Customer has not responded or provided something needed
Waiting for third party	An external vendor or partner is holding things up
Internal dependency	Another task in this or another project must finish first
Other	Anything else — requires a text description

How to flag a sub-task:

1. A small flag icon appears on each sub-task row (next to edit/delete icons on hover).
2. Clicking the flag icon opens a small modal:
 - Flag type (dropdown of values above, required)
 - Blocking reason (free text, required for "Other", optional but shown for all types)

- Waiting on (free text — name of a person, team, system, or vendor)
3. On save, the sub-task status is automatically set to "Blocked" (or "Awaiting Feedback" if the type is "Waiting for approval" or "Waiting for customer").
 4. The flag icon on the row changes to a filled/coloured icon to indicate a flag is active.
 5. Hovering over the filled flag icon shows a tooltip with the flag summary.

Removing a flag:

- Clicking the filled flag icon on a flagged sub-task opens the same modal pre-filled.
- A "Clear flag" button at the bottom of the modal removes the flag and sets the status back to "In Progress".

Data changes:

Sub-task object gains: `flag` object with fields: `type`, `reason`, `waitingOn`.

Visibility of flags:

- In the task panel, flagged sub-tasks show the filled flag icon.
- In the activity group header, if any sub-task in the group is flagged, a small flag count appears (e.g., "  2") next to the group's progress bar.
- On the Dashboard "Needs Attention" panel, any project with flagged sub-tasks gets an indicator even if the project-level status is "On Track".
- On the "My Tasks" page, flagged tasks are highlighted with the flag icon and sorted to the top of the list (within their overdue/due-soon groupings).

2.5 Data Model (updated)

Project (*no change*)

Task group (*no change*)

Sub-task (updated):

```

{
  name: string,
  done: boolean,           // kept for backward compat, synced with status
  due: string,             // date string YYYY-MM-DD
  assignee: string,        // name of assignee, or empty string
  status: string,          // one of the 6 status values
  flag: {                  // null if not flagged
    type: string,
    reason: string,
    waitingOn: string
  } | null
}

```

New top-level state:

```

teamMembers: string[]      // ["Rahul", "Priya", "Amit", "Sneha", ...]

```

2.6 UI Changes Summary

Location	Change
Sidebar	Add "My Tasks" nav item; add gear icon in footer for team management
Project row (collapsed)	No change
Task panel header	Add "+ Add phase" button
Activity group header	Add edit and delete icons (visible on hover); add flag count indicator
Sub-task row	Add assignee chip, status badge, flag icon, edit icon, delete icon; rearrange layout
Sub-task row — add mode	Inline input row at bottom of group
Dashboard — Needs Attention	Add blocked/flagged sub-task indicator per project
New: My Tasks page	Full new page with person picker and task list
New: Flag modal	Small modal for entering flag details
New: Team modal	Modal for managing team members

2.7 Status and Badge Colour Reference

Status	Badge class (existing pattern)	Dot colour
Not Started	badge-gray	var(--text3)
In Progress	badge-blue	var(--accent)
In Testing	badge-teal	var(--teal)
Awaiting Feedback	badge-amber	var(--amber)
Blocked	badge-red	var(--red)
Done	badge-green	var(--green)

PART 3: FRONTEND TASK BREAKDOWN

Actionable tasks for a developer to implement the features above. Each task is self-contained and can be picked up independently unless dependencies are noted.

Phase 0 — Data Migration (no UI, do first)

#	Task	Notes
0.1	Update <code>mkTasks()</code> to generate sub-tasks with new fields	Add <code>assignee: ""</code> , <code>status: "Not Started"</code> , <code>flag: null</code> to every sub-task
0.2	Add <code>teamMembers</code> array to top-level state	Pre-populate from existing owner names
0.3	Update <code>taskProg()</code> to count by status "Done" instead of <code>done === true</code>	Keeps progress bars accurate
0.4	Update <code>toggleSub()</code> to sync <code>done</code> and <code>status</code> fields together	Checking → status = "Done"; unchecking → status = "In Progress"

Phase 1 — Sub-task Row Redesign

#	Task	Depends on	Notes
1.1	Redesign <code>sub-row</code> CSS grid	0.1	New columns: name, assignee chip, date, status badge, checkbox, action icons
1.2	Render assignee chip on sub-task row	0.2	Show initials avatar; clicking opens assignee picker dropdown
1.3	Build assignee picker dropdown	0.2	Small popover listing team members + "Unassign" option; closes on click-outside
1.4	Render status badge on sub-task row	0.1	Clickable; opens status picker dropdown
1.5	Build status picker dropdown	0.1, 0.4	Lists all 6 statuses; on pick, updates status and syncs done field
1.6	Show edit + delete icons on hover	—	Use <code>onmouseenter/leave</code> on the row; icons styled as small <code>btn-sm</code> buttons
1.7	Inline edit for sub-task name	1.6	Replace name span with input on edit icon click; Enter saves, Escape cancels
1.8	Inline delete confirmation for sub-task	1.6	Replace row content with "Delete? Yes / Cancel" inline; on Yes remove from array

Phase 2 — Add / Edit / Delete Activity Groups

#	Task	Depends on	Notes
2.1	Add "+ Add step" link at bottom of each group's sub-task list	Phase 1	Renders an inline input row; Enter saves new sub-task with defaults
2.2	Add "+ Add phase" button in task panel header	—	Opens a small inline input or simple prompt; saves new empty group

#	Task	Depends on	Notes
2.3	Add edit icon on activity group header	—	Inline input replaces group name text; Enter saves
2.4	Add delete icon on activity group header	—	Inline confirmation; on confirm removes group and all its sub-tasks
2.5	Validate non-blank names	2.1, 2.3	Show red inline message "Name cannot be blank" if user tries to save empty

Phase 3 — Flag / Block System

#	Task	Depends on	Notes
3.1	Add flag icon to sub-task row	Phase 1	Outline icon = no flag; filled/red = flag active
3.2	Build flag modal HTML + CSS	—	Fields: flag type dropdown, reason textarea, waitingOn input, Save + Clear flag + Cancel buttons
3.3	<code>openFlagModal(pid, gi, si)</code> function	3.2	Pre-fills if flag already set; shows "Clear flag" only if flag exists
3.4	<code>saveFlag()</code> function	3.3	Saves flag object to sub-task; sets status to "Blocked" or "Awaiting Feedback" based on flag type
3.5	<code>clearFlag()</code> function	3.3	Removes flag object; sets status to "In Progress"; re-renders
3.6	Flag tooltip on hover	3.1	Use a CSS <code>title</code> attribute or a small custom tooltip div showing flag type + reason
3.7	Flag count indicator on group header	3.1	Count sub-tasks in group where <code>flag !== null</code> ; show "  N" if N > 0
3.8	Flagged project indicator on Dashboard	3.1	In Needs Attention panel, count flagged sub-tasks per project; append badge

Phase 4 — My Tasks Page

#	Task	Depends on	Notes
4.1	Add "My Tasks" nav item to sidebar	—	Between Customers and sidebar footer
4.2	Add <code>currentPage === "myTasks"</code> branch to <code>render()</code>	4.1	Calls <code>renderMyTasks()</code>
4.3	Build person picker control	0.2	Tab strip or dropdown of <code>teamMembers</code> ; selection stored in <code>selectedMember</code> state var
4.4	Build <code>renderMyTasks()</code> function	4.3	Iterates all projects and all sub-tasks; filters where <code>assignee === selectedMember</code>
4.5	Sort and group tasks in My Tasks	4.4	Groups: Overdue (past due, not Done), Due Soon (within 7 days), Upcoming, No date, Done
4.6	Render each task row in My Tasks	4.4	Show: project name, group name, step name, status badge, due date, flag icon if flagged
4.7	Make project name a link	4.6	Clicking navigates to Projects page with that project expanded

Phase 5 — Team Management

#	Task	Depends on	Notes
5.1	Add gear icon to sidebar footer	—	Opens team management modal
5.2	Build team management modal	—	Lists current members; "+ Add member" inline input; remove button per member
5.3	Guard remove for assigned members	5.2	If a member has any sub-tasks assigned, disable their remove button with tooltip

#	Task	Depends on	Notes
5.4	Sync team list with assignee pickers	5.2	Assignee dropdowns (Phase 1.3) pull from <code>teamMembers</code> array dynamically

Phase 6 — Dashboard Updates

#	Task	Depends on	Notes
6.1	Update Needs Attention to include flagged projects	3.8	Show even if project status is "On Track" if it has flagged steps
6.2	Add "Blocked steps" count to Needs Attention rows	6.1	Small red badge showing total blocked steps per project in the list

PART 4: UX NOTES AND DESIGN GUIDANCE

For whoever implements the UI — these notes ensure the new features feel consistent with the existing design.

4.1 Interaction Principles (match existing app)

- All icons, dropdowns, and action buttons in the task panel should use the same size scale as existing elements (font-size 11-13px, btn-sm padding).
- Hover states: use `var(--surface2)` background on sub-task rows, same as today.
- Modals: use the existing `.modal-bg` + `.modal` pattern. New modals (flag, team) should match width and padding.
- Colour usage: do not introduce new colours. Use the existing CSS variables (`--green`, `--amber`, `--red`, `--accent`, `--teal`, `--purple`, `--text2`, `--text3`).
- Destructive actions (delete) must always require a confirmation step — never delete immediately on first click.

4.2 Sub-task Row Layout (target)

The row should use a CSS grid with these approximate column widths:

```
[name - flex 1] [assignee - 90px] [date - 120px] [status - 110px] [checkbox - 30px] [icons - 54px]
```

On narrower panels, the icons column collapses to show only on hover of the full row.

4.3 Status Badge Behaviour

- The badge should look clickable (cursor: pointer, slight hover background change).
 - The dropdown that appears should be a small absolute-positioned div (not a native `<select>`) so it can be styled consistently.
 - The dropdown closes when the user clicks outside it (`document.addEventListener("click", ...)` pattern already used in the codebase).
-

4.4 Flag Icon Design

- Use a flag emoji () or an SVG flag icon for consistency.
 - Unflagged state: faint gray outline version of the icon (opacity ~30%).
 - Flagged state: filled red.
 - The icon should be 14-16px and vertically centred in the row.
-

4.5 My Tasks Page Layout

- Person picker at the top: a tab strip works well for up to 6-8 people. If team grows larger, switch to a dropdown.
 - Each project section in the task list should use the same `.activity-group` card style as the task panel.
 - Overdue items get the sub-task name displayed in `var(--red)` colour.
 - Done items get a strikethrough and reduced opacity (same as existing done style).
-

4.6 Empty States

Each new feature needs an empty state:

Scenario	Message
No sub-tasks in a group	"No steps yet. Click '+ Add step' to add one."
No flagged tasks anywhere	(no special state needed — flags are per-task)
My Tasks — no tasks for selected person	"No tasks assigned to [name] yet."
My Tasks — no team members	"Add team members using the  icon in the sidebar."

PART 5: OPEN QUESTIONS

Decisions needed before or during implementation.

#	Question	Options	Recommended default
Q1	Should the existing project-level "owner" field remain, or be replaced by per-task assignees?	Keep both / Replace with assignees	Keep both — owner = accountable person; assignee = doing the work
Q2	Should "Done" sub-tasks still move the progress bar, or should we weight by status?	Binary (done/not done) / Weighted	Keep binary for simplicity
Q3	Can one sub-task have multiple assignees?	Single / Multiple	Single for now — simpler UI
Q4	Should status changes be logged with a timestamp (an audit trail)?	Yes / No	No for v2 — adds complexity
Q5	Should the My Tasks page show tasks from all projects or only active (non-Completed) ones?	All / Active only	Active only by default, with a "Show completed projects" toggle
Q6	Should adding a new activity group from the task panel persist across page re-renders?	In-memory only (yes, like everything else) / Save to localStorage	In-memory, consistent with current app behaviour

End of document.